

- Cutting Planes / Branch and Bound -

Exercise 1: Cutting planes (Branch and Cut)

Solve the following problem using a branch and bound algorithm:

$$(P_1) \begin{cases} \max Z_1 = x_1 + 5x_2 \\ x_1 + 10x_2 \leq 20 \\ x_1 \leq 2 \end{cases} \quad \begin{matrix} x_1 \in \mathbb{N} \\ x_2 \in \mathbb{N} \end{matrix}$$

Exercise 2: Branch and Bound for the Assignment Problem

Consider the following assignment problem, where $n = 4$ and the cost matrix $C = [c_{ij}]$ is given as follows:

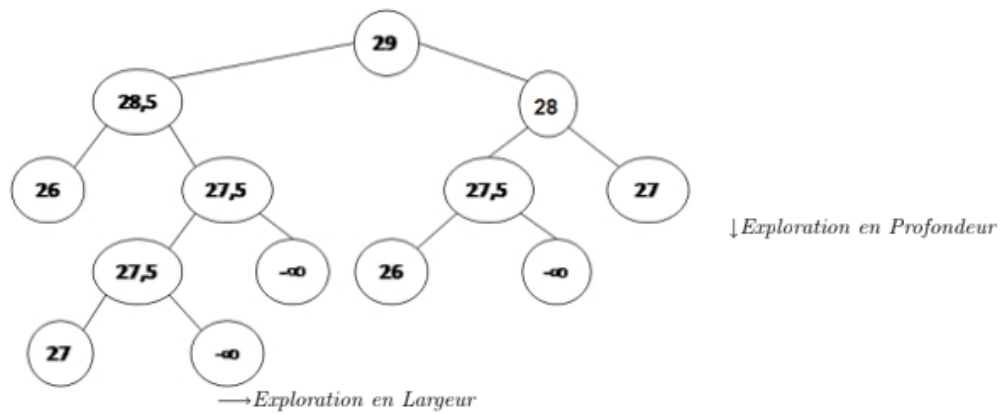
$$C = \begin{bmatrix} 9 & 2 & 7 & 8 \\ 6 & 4 & 3 & 7 \\ 5 & 8 & 1 & 8 \\ 7 & 6 & 9 & 4 \end{bmatrix}$$

Each row i represents a worker and each column j represents a job. The entry c_{ij} denotes the cost incurred if worker i is assigned to job j .

- (1) Apply the Hungarian algorithm to find the optimal assignment that minimizes the total cost.
- (2) Apply a greedy heuristic and compare the result with the optimal solution.
- (3) Use the Branch and Bound method to solve this assignment problem.
- (4) Compare the results obtained from the three methods. Discuss:
 - The optimality of each method.
 - The computational complexity.
 - The advantages and limitations of each method in practical applications.

Exercise 3: Analysis of a Branch-and-Bound Search Tree

We consider the following search tree associated with a branch-and-bound method, where the circled numbers indicate evaluations, and we assume that all nodes are fathomed.



- (1) Is it a minimization problem or a maximization problem?
- (2) Which of the two optima is reached first if a depth-first search is applied?
- (3) Which of the two optima is reached first if a breadth-first search is applied?